BITS PILANI, DUBAI CAMPUS
SECOND SEMESTER 2025 – 2026
Department of Computer Science

Course Code: CS F429 | **Max Marks: 40**
Course Title: Natural Language Processing | **Weightage: 40%**
Mode of Assessment: Open Book | **Nature: Team-Based**

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Midsem Evaluation Rubric — Research Results & Experiments

For the midsem evaluation, each team must run all specified experiments, fill every result table with real numbers from the held-out test set, and demonstrate a live pipeline. Every mark is determined by an objective criterion the AUROC your composite achieves on the test set, how early your metric signal peaks relative to the hallucinated token, and whether your pipeline runs on an input the evaluator provides on the spot.

> The test set must be treated as unseen until final evaluation re-running experiments after observing test results or reporting validation numbers in place of test numbers inflates performance and invalidates the comparison against baselines and SOTA. Every table cell must contain a real number not a placeholder, not an estimate. The code and demo component will be assessed live on an input you have not seen before, and any team member may be asked to explain any part of the implementation without notice.

Marks summary

Component	Track A max	Track B max	What determines the mark
Experiments 1 & 2 — Composite AUROC on RAGTruth	4	4	AUROC band of final composite on test set
Experiment 3 — Temporal precedence	2	2	Earliest position at which signal peaks
Experiment 4 — Cross-domain transfer (HaluEval)	2	2	AUROC on HaluEval + explanation quality
Experiment 5 — Type breakdown / causal patching	2	2	AUROC gap magnitude + mechanistic explanation
Experiments 6–8 — Deeper analysis	2	2	Completeness + SOTA gap closed
Code & demo	3	3	Runs on unseen input + member explains code
Total	15	15	

Track A and Track B use different AUROC thresholds because hidden-state extraction (Track B) requires substantially more infrastructure than token probability extraction (Track A). The AUROC target for 4 marks is 0.75 on Track A and 0.70 on Track B. All other components are marked identically.

Experiments shared across both tracks

The following experiments use the same protocol and the same marking criteria for both tracks. The metrics and numbers differ — the structure is identical.

- Experiment 3: temporal precedence (t−3 to t+1 relative to first hallucinated token)
- Experiment 4: cross-domain transfer to HaluEval — zero-shot, no re-fitting
- Code & demo: live pipeline on evaluator-provided unseen input

Track A — Dynamic Uncertainty-Aware Attribution

Five experiments grouped into four assessed components totalling 15 marks.

Experiments 1 & 2 Composite AUROC on RAGTruth 4 marks

Run all 5 metrics individually (E1), then build the composite incrementally (E2). The mark is determined by the AUROC of your final composite on the test set. Fill the table below and submit with bootstrap CI.

Metric / Composite	AUROC ↑	F1 span ↑	Spearman ρ ↑	ECE ↓
Entropy only (Baseline 1)	___	___	___	___
SelfCheckGPT (Baseline 2)	___	___	___	___
+ Information Gain	___	___	___	___
+ KL divergence	___	___	___	___
+ Confidence drop	___	___	___	___
+ Semantic entropy	___	___	___	___
Full composite	___	___	___	___

Marks allocation

Marks	Condition
4	Composite AUROC ≥ 0.75 on test set, bootstrap CI reported, both baselines beaten
3	Composite AUROC 0.68–0.74, both baselines beaten
2	Composite AUROC 0.60–0.67, at least one baseline beaten
1	Composite AUROC 0.52–0.59, or only individual metrics without composite
0	AUROC < 0.52, or results from validation set, or tables not filled

SOTA reference: SelfCheckGPT ~0.65, Semantic Entropy ~0.70, LUMINA supervised ~0.87. Target of ≥ 0.75 unsupervised is competitive with the best published unsupervised methods.

Experiment 3 Temporal precedence *[both tracks]* 2 marks

Report mean metric score at positions t−3 to t+1 relative to the first hallucinated token t. Include a line plot in the report. The mark depends on how early the signal peaks.

Position	IG	KL	Predictive H	Semantic H	Conf. drop
t−3	___	___	___	___	___
t−2	___	___	___	___	___
t−1	___	___	___	___	___
t (onset)	___	___	___	___	___
t+1	___	___	___	___	___

Marks allocation

Marks	Condition
2	Any metric peaks at t−2 or earlier; Mann-Whitney U test reported; line plot included
1	Best metric peaks at t−1; or peaks earlier but statistical test missing
0	All metrics peak at t or later; or table filled without plot or significance test

A metric peaking at t−2 or earlier is a strong pre-generation signal and constitutes a genuine contribution.

Experiment 4 Cross-domain transfer — HaluEval zero-shot *[both tracks]* 2 marks

Run full composite on HaluEval QA subset. No re-training or re-calibration. Mark is based on the AUROC achieved and the quality of explanation for any rank instability observed.

Metric	AUROC RAGTruth	AUROC HaluEval	Drop	Rank stable?
Full composite	___	___	___	___
Information Gain	___	___	___	___
KL divergence	___	___	___	___
Semantic entropy	___	___	___	___

Marks allocation

Marks	Condition
2	HaluEval composite AUROC ≥ 0.68 AND rank instability explained mechanistically
1	HaluEval AUROC 0.58–0.67; or ≥ 0.68 but instability not explained
0	HaluEval AUROC < 0.58; or experiment not attempted

Zero-shot transfer AUROC will typically drop 5–10 points from RAGTruth. A drop > 15 points must be explained.

Experiment 5 Hallucination type breakdown 2 marks

RAGTruth labels each hallucination as contradictory, unsupported, or fabricated. Report composite AUROC per type. The mark depends on the magnitude of variation found and the depth of explanation.

Hallucination type	Count in test set	Composite AUROC	Best single metric
Contradictory	___	___	___
Unsupported	___	___	___
Fabricated	___	___	___

Marks allocation

Marks	Condition
2	AUROC gap > 0.10 across types AND mechanistic explanation of why the gap exists
1	Table filled with real numbers but gap unexplained; or gap ≤ 0.10
0	Table not filled or all values identical

Expected finding: 'contradictory' hallucinations have higher AUROC than 'unsupported' because entropy shifts more when the model contradicts retrieved context than when it simply omits it.

Experiments 6–8 Generator analysis, failure modes, SOTA gap 2 marks

E6: AUROC by generator model (Llama-2-7B, 13B, GPT-3.5, GPT-4). E7: at least 3 failure cases with mechanistic explanation. E8: SOTA gap table with LUMINA (~0.87) and ReDeEP (~0.82) as upper-bound rows, and structured gap analysis paragraph.

Marks allocation

Marks	Condition
2	All three attempted: generator table filled + ≥ 3 failure cases explained + SOTA gap closed ≥ 50% over entropy-only baseline
1	At least one of E6/E7/E8 attempted with substantive content; or all three attempted but shallowly
0	None attempted or only SOTA table without gap analysis

SOTA gap closed = (your AUROC – entropy baseline AUROC) / (LUMINA AUROC – entropy baseline AUROC) × 100%. Closing ≥ 50% means: if entropy baseline is 0.61 and LUMINA is 0.87, your composite should reach ≥ 0.74.

Code & Demo Live pipeline demonstration 3 marks

During the midsem session the evaluator will provide an unseen passage with retrieved context. Your pipeline must accept it as input, run end-to-end, and produce token-level metric scores. Any team member may be asked to explain any part of the code.

Marks allocation

Marks	Condition
3	Pipeline runs on evaluator's unseen input without errors; any team member can explain the metric computation when asked
2	Pipeline runs correctly but only on team's own prepared inputs; or runs on unseen input but no member can explain the code
1	Pipeline partially runs — metric computation starts but fails before producing output
0	Pipeline does not run

The evaluator will ask at least one team member: 'Explain exactly how KL divergence is computed in your code, and show me where in the code it happens.' This question will be asked without warning.

Track B — Pre-Generation Causal Drift Metric

Five experiments grouped into four assessed components totalling 15 marks. AUROC thresholds are lower than Track A to reflect the additional infrastructure required for hidden-state extraction and causal patching.

Experiments 1 & 2 Composite AUROC on RAGTruth + layer localisation 4 marks

Run all 5 representation metrics individually (E1) and identify which layer range carries the signal (E2). The mark is determined by your composite AUROC on the test set. μ_l and $\sum_l$ must be estimated on training split only.

Metric / Composite	AUROC ↑	F1 span ↑	Spearman ρ ↑	ECE ↓
Attention entropy (Baseline 1)	___	___	___	___
Logit confidence (Baseline 2)	___	___	___	___
Cosine drift	___	___	___	___
Mahalanobis distance	___	___	___	___
Logit lens divergence	___	___	___	___
PCA deviation	___	___	___	___
CIE top-3 layers	___	___	___	___
Full composite	___	___	___	___

Marks allocation

Marks	Condition
4	Composite AUROC ≥ 0.70 on test set; layer-profile plot included; both baselines beaten
3	Composite AUROC 0.63–0.69; both baselines beaten; layer plot present
2	Composite AUROC 0.55–0.62; or ≥ 0.63 but no layer analysis
1	Composite AUROC 0.48–0.54; or individual metrics only without composite
0	AUROC < 0.48; or results from validation set; or no leakage check reported

Track B thresholds are lower than Track A because extracting and computing on hidden states is substantially harder infrastructure work. SOTA reference: ReDeEP ~0.82, probabilistic distance method ~0.80 unsupervised.

Experiment 3 Causal intervention — activation patching 2 marks

Patch hidden states in both directions: faithful→hallucinated and hallucinated→faithful. Report CIE by component type. Minimum 50 patching experiments. Discuss agreement or disagreement with ReDeEP's Knowledge FFN / Copying Head finding.

Component	CIE faithful→halluc.	CIE halluc.→faithful	Critical?
Early attn heads (1–25%)	___	___	___
Mid FFN layers (26–75%)	___	___	___

Late FFN layers (76–100%)	___	___	___
Copying heads (last 25%)	___	___	___

Marks allocation

Marks	Condition
2	Both patching directions reported; CIE significant in ≥ 2 components ($p < 0.05$); ReDeEP comparison written
1	Only one patching direction; or CIE computed but no significance test; or ReDeEP not discussed
0	Patching not attempted; or fewer than 50 experiments

ReDeEP finds that late FFN layers are the critical component. If your results agree, explain the mechanism. If they disagree, that is equally valuable — explain why.

Experiment 4 Temporal precedence *[both tracks]* 2 marks

Report mean drift at t−3 to t+1 for all 5 representation metrics. The causal claim of Track B is that drift rises before t. Include a line plot.

Position	Cosine drift	Mahalanobis	Logit lens	PCA dev.	CIE
t−3	___	___	___	___	___
t−2	___	___	___	___	___
t−1	___	___	___	___	___
t (onset)	___	___	___	___	___
t+1	___	___	___	___	___

Marks allocation

Marks	Condition
2	Any metric peaks at t−2 or earlier; Mann-Whitney U reported; line plot included
1	Best metric peaks at t−1; or peaks earlier but without significance test
0	All metrics peak at t or later; or table filled without plot

Same structure as Track A E3. Cosine drift and Logit lens divergence are most likely to show early precedence based on existing mechanistic interpretability literature.

Experiment 5 Cross-domain transfer — HaluEval zero-shot *[both tracks]* 2 marks

Run composite on HaluEval with no re-fitting of μ_l, Σ_l, or PCA components. Identify which metric degrades most and explain why.

Metric	AUROC RAGTruth	AUROC HaluEval	Drop
Cosine drift	___	___	___
Mahalanobis distance	___	___	___
Logit lens divergence	___	___	___
PCA deviation	___	___	___
Full composite	___	___	___

Marks allocation

Marks	Condition
2	HaluEval composite AUROC ≥ 0.62 AND the most brittle metric identified and explained
1	HaluEval AUROC 0.53–0.61; or ≥ 0.62 but brittleness unexplained
0	HaluEval AUROC < 0.53; or experiment not attempted

Track B zero-shot threshold is 0.62 vs Track A's 0.68 because re-fitting $\sum_l$ across domains is expensive and the distribution shift is harder to handle in representation space.

Experiments 6–8 FFN decomposition, failure modes, SOTA gap 2 marks

E6: separately compute drift on FFN output vs attention output per layer range — state whether signal localises to FFN or attention. E7: at least 3 failure cases with mechanistic explanation. E8: SOTA gap table with ReDeEP (~0.82) and LUMINA (~0.87) as upper bounds.

Marks allocation

Marks	Condition
2	All three attempted: FFN vs attention table filled + ≥ 3 failure cases explained + SOTA gap closed ≥ 50% over attention entropy baseline
1	At least one of E6/E7/E8 with substantive content; or all three attempted but shallowly
0	None attempted; or SOTA table without analysis

E6 expected finding: hallucination signal concentrates in mid-to-late FFN layers (consistent with ReDeEP). If your FFN/attention decomposition contradicts this, that is a publishable finding — explain it thoroughly.

Code & Demo Live pipeline demonstration 3 marks

The evaluator will provide an unseen passage with retrieved context. Your pipeline must extract hidden states, compute all 5 representation metrics, and output token-level scores. Any team member may be asked to explain the code.

Marks allocation

Marks	Condition
3	Pipeline runs on evaluator's unseen input; hidden state extraction working; any team member can explain the metric computation
2	Pipeline runs on prepared inputs only; or runs on unseen input but member cannot explain the code
1	Hidden state extraction works but metric computation fails before producing output
0	Pipeline does not run

The evaluator will ask at least one member: 'Show me where in the code Mahalanobis distance is computed, and explain what μ_i and $\sum_i$ are in your implementation.' This question will be asked without warning.